

Algebra 1  
Unit 2  
Lesson 2 Practice

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

1.  $\frac{3x^6y^5}{12xy^8}$

$$\frac{1x^5}{4y^3} = \frac{x^5}{4y^3}$$

2.  $\frac{2a^2b^7}{12a^3b^4}$

$$\frac{1b^3}{6a} = \frac{b^3}{6a}$$

3.  $\left(\frac{-2s^3t^6}{8t^6}\right)^5$

$$\frac{-32s^{15}}{32768} = \frac{-s^{15}}{1024}$$

4.  $\frac{-2k^5j^9}{-6k^4j^{10}}$

$$\frac{1k}{3j} = \frac{k}{3j}$$

5.  $\frac{28w^8z^6}{42w^{10}z^4}$

$$\frac{2z^2}{3w^2}$$

$$6. \quad \left(\frac{xy^9}{-5y^4}\right)^0$$

1

$$7. \quad \frac{6r^8s^5}{-12r^5s^8}$$

$$\frac{-1r^3}{2s^3} = \frac{-r^3}{2s^3}$$

$$8. \quad \frac{5x^5}{60x^{11}s^3}$$

$$\frac{1}{12x^6s^3}$$

$$9. \quad \left(\frac{13x^0z^5}{-x^4z}\right)^2$$

$$\frac{169 \cdot 1 \cdot z^{10}}{x^8z^2} = \frac{169z^8}{x^8}$$

$$10. \quad \left(\frac{-25r^{10}p^8}{-10r^4z}\right)^3$$

$$\frac{125r^{18}p^{24}}{8z^3}$$